and the y-intercept c as:

```
c=y -mx                                    (3)
```

where x and y are the average values of x and y variables, respectively.

To illustrate the implementation of linear regression, let us consider an example of a simple relationship between a car's age and its price. The relationship is depicted in a tabular form as shown below:

Car age (years)	Price (Rs)
4	630,000
4	580,000
5	570,000
5	450,000
7	450,000
7	420,000
8	410,000
9	310,000
10	270,000
11	250,000
12	220,000

Table 1: The relation between a car's age and its price (carprice)

A table *carprice* has been created with a database dataanalysis, and used in an SQL query to determine regression analysis coefficients.

The task is to find the regression coefficients m and c using the formula given above. This can be done with other popular programming languages like R and Python, conveniently, through a language-specific RDBMS table import mechanism. With SQL, one can implement this directly within the database platform itself.

Implementation

SQL implementation of regression analysis is a straightforward exercise because the entire mathematical formulation can be implemented directly within the query. The query structure is a little cumbersome and requires attention. It need the sum of products, of different combinations. So, at the outset, let us try the slope alone. The slope variable m is the ratio of two sums of products of two different terms, and can be written in an SQL query as follows:

```
SELECT ROUND(
       SUM(
(Age-(SELECT SUM(Age)/ COUNT(Age) AS avgAge FROM carprice))*
(Price-(SELECT SUM(Price)/ COUNT(Price) AS avgPrice FROM
carprice))
          )
/ SUM(
(Age-( SELECT SUM(Age)/ COUNT(Age) AS avgAge FROM carprice)) *
(Age-( SELECT SUM(Age)/ COUNT(Age) AS avgAge FROM carprice))
) ,2) AS m
FROM
  carprice AS a;
```

If it is clear to you, you may now try the y-intercept c. It has three terms: y, x and m. Using the query expression of m, as stated in the previous query, we can write the following SQL statement for Equation 3.

```
SELECT  round(
        (SUM(Price)/ COUNT(Price)) -     #y
        (SUM                             #m
         (
(Age-(SELECT SUM(Age)/ COUNT(Age) AS avgAge FROM carprice))*
(Price-(SELECT SUM(Price)/ COUNT(Price) AS avgPrice FROM
carprice))
) /
     SUM
          (

(Age-(SELECT SUM(Age)/ COUNT(Age) AS avgAge FROM carprice)) *
(Age-(SELECT SUM(Age)/ COUNT(Age) AS avgAge FROM carprice))
     )


            )*

(SUM(Age)/ COUNT(Age)),2) as c        #x
  FROM
        carprice AS test
```

The first query will return m as -0.50 and the second query will give c as 7.84. A negative slope value indicates lower values of the price with an increase in the age of the car. With the help of these two coefficients, one can estimate the selling price of a car if its age is given. For example, if the age of a car in the garage is three years, then the approximate price is:

```
SELECT -0.50*3+7.84      →   6.84 lakhs.
```